

2 Two Motivating Problems

2.1 Time-Series Prediction

Consider the autonomous discrete-time dynamical system

$$x_{t+1} = f(x_t), \quad x_0 = x_{\text{init}}, \quad (2.1)$$

which we will often write more compactly as

$$x_+ = f(x). \quad (2.2)$$

Here $x \in \mathbb{R}^n$ is the system state, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defines the system dynamics.

Suppose that f is unknown, but we are given a data set of trajectories

$$D = \left\{ x_t^{(i)} \right\}_{t=0, \dots, T}^{i=1, \dots, N} \quad (2.3)$$

generated by drawing N initial conditions $x_0^{(i)} \stackrel{\text{iid}}{\sim} P_0$ and observing the evolution of (2.2).

Our goal is to learn a dynamics map $\hat{f}$ such that

- (a) $\hat{f} \approx f$ in some appropriate sense; and
- (b) $\hat{x}_t \approx x_t^*$ for $t = 0, \dots, T$ (in expectation, or with high probability), where

$$\hat{x}_{t+1} = \hat{f}(\hat{x}_t), \quad x_{t+1}^* = f(x_t^*), \quad (2.4)$$

and $\hat{x}_0 = x_0^* = x_{\text{init}} \sim P_0$.

Remark 2.1. Here x_{init} is itself understood to be a draw from P_0 in the same way the N training initial conditions were.

Why this is challenging: we learn $\hat{f}$ using the true system data $\{x_t^*\}$, but generate predictions $\hat{x}$ *auto-regressively* (i.e., via self-feedback).

Key Question 2.1. When does $\hat{f} \approx f$ on the true data distribution imply $\hat{x}_t \approx x_t^*$ for $\hat{x}_0 = x_0^* = x_{\text{init}} \sim P_0$?

2.2 Imitation Learning

Consider a discrete-time dynamical system

$$x_+ = f(x, u), \quad x_0 = x_{\text{init}}, \quad (2.5)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control (actuation) input, $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ are the system dynamics, and x_{init} is the known initial condition.

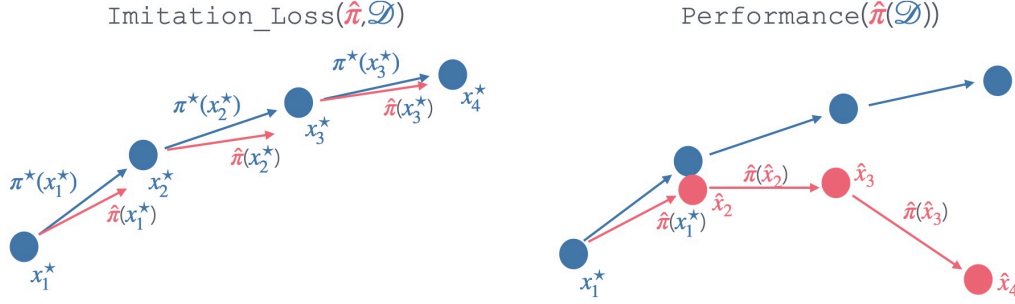


Figure 2.1: Left: $\hat{\pi}$ is learned on expert data $\{x_t^*\}$ generated by f^{π^*} . Right: at deployment, $\hat{\pi}$ encounters states $\{\hat{x}_t\}$ generated by $f^{\hat{\pi}}$, which may be far away from the training distribution of $\{x_t^*\}$.

A *Markovian feedback policy* π is a map $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defining the control input $u = \pi(x)$.¹ Define the *closed loop system* f^π under a Markovian policy π as

$$f^\pi(x, v) := f(x, \pi(x) + v), \quad (2.6)$$

i.e., $f^\pi(x, v)$ is obtained by setting $u = \pi(x) + v$ in (2.5). Here v is a possible auxiliary input (or error); we write $f^\pi(x) := f^\pi(x, 0)$ for the base closed loop map with no auxiliary input.

Suppose that there is an expert policy π^* for which we are given a data set of expert state/action pairs

$$D = \left\{ \left(x_t^{*,(i)}, \pi^*(x_t^{*,(i)}) \right) \right\}_{t=0, \dots, T-1}^{i=1, \dots, N}, \quad (2.7)$$

each generated i.i.d. by drawing initial conditions $x_0^{*,(i)} \stackrel{\text{iid}}{\sim} P_0$ from some distribution P_0 , and then rolling out

$$x_{t+1}^* = f^{\pi^*}(x_t^*) = f(x_t^*, \pi^*(x_t^*)), \quad (2.8)$$

for $t = 0, \dots, T-1$.

Our objective is to learn a policy $\hat{\pi}$ using the data set D so that

- (a) $\hat{\pi} \approx \pi^*$ in some appropriate sense; and
- (b) $\hat{x}_t \approx x_t^*$ for $t = 0, \dots, T$ (in expectation, or with high probability), where

$$\hat{x}_{t+1} = f^{\hat{\pi}}(\hat{x}_t) = f(\hat{x}_t, \hat{\pi}(\hat{x}_t)), \quad x_{t+1}^* = f^{\pi^*}(x_t^*) = f(x_t^*, \pi^*(x_t^*)), \quad (2.9)$$

and $\hat{x}_0 = x_0^* = x_{\text{init}} \sim P_0$.

Key Question 2.2. When does $\hat{\pi} \approx \pi^*$ on the expert data distribution imply that $\hat{x}_t \approx x_t^*$ for $\hat{x}_0 = x_0^* = x_{\text{init}} \sim P_0$?

Why this is challenging: we learn $\hat{\pi}$ using the expert system data $\{x_t^*\}$, but at test time see states $\{\hat{x}_t\}$ generated by the learned closed loop system $f^{\hat{\pi}}$. These may look very different from our training data, leading to compounding error, see Fig. 2.1.

¹We assume throughout that everything is time invariant, unless otherwise specified.

2.3 Common Questions

The problem statements above lead to a surprisingly rich set of questions:

1. When is this even possible?
2. When can it be done with a reasonable amount of data?
3. How should we parameterize what we're learning ($\hat{f}$ in time-series prediction, $\hat{\pi}$ in imitation learning)?
4. How should we collect the data (the distribution P_0 over initial conditions, and, in imitation learning, the expert policy itself)?
5. What properties should the true dynamics f satisfy to make this easier?
6. What properties should the closed loop under the true/expert behavior (f itself in time-series prediction, f^{π^*} in imitation learning) satisfy to make this easier?
7. What are the right notions of closeness – between the learned and true objects ($\hat{f} \approx f$, or $\hat{\pi} \approx \pi^*$), and between the resulting trajectories?

2.4 Key Components of Our Answers

1. **Feedback is capricious!** Both problems above are feedback systems: time-series prediction's autoregressive rollout $\hat{x}_{t+1} = \hat{f}(\hat{x}_t)$ feeds each prediction back in to generate the next, and imitation learning's control law $u = \pi(x)$ feeds the current state back in as the next action. We've seen that feedback can help make systems robust to uncertainty, but in doing so can also induce unexpected fragilities. We'll see some of these fragilities and how they manifest in both settings, and how to avoid them by harnessing feedback.
2. **(Robust) stability is key!** We'll see that traditional notions of stability (e.g., to an equilibrium point) remain crucial, but we'll also need to understand stability at the *trajectory* level, i.e., when do the trajectories of “similar” dynamical system pairs (e.g. $(f, \hat{f})$, or $(f^{\pi^*}, f^{\hat{\pi}})$) stay close to each other?